FU Xiao

(+86)19858125696 | lemonaddie0909@zju.edu.cn | https://fuxiao0719.github.io/

Education

Zhejiang University (ZJU) Hangzhou, China
Bachelor of Infomation Eng. | College of Information Science&Electronic Engineering (ISEE) *Sept. 2018 – Jul. 2022*

- **GPA:** 3.96/4.00, 90.63/100, **Ranking:** 1/144
- **Laureate of National Scholarship (twice), Graduated with Dean's Honor**
- **Honors Program:** ZJU Morningside Cultural China Scholars Program (36 students shortlisted from 12,000 freshmen with distinguished innovative capability and leadership)
- **Language Proficiency:** TOEFL(iBT): 99(R:27,L:25,S:22,W:25)

Research Interests

3D Computer Vision, in particular neural scene representations (the way neural networks learn to represent information on our world), which consists of several aspects (1) High-quality 3D reconstruction of real-world scenes from sparse inputs (2) Efficient and photo-realistic image synthesis and generation of large-scale real-world scenes (3) Convex optimization.

Publications

∗ **equal contribution, † corresponding author**

- Xiao Fu*, Shangzhan Zhang*, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao†. "**Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation**". *2022 International Conference on 3D Vision (3DV)*, Prague, Czechia, pp. 301-311 [**HomePage**|**Paper**|**Poster**|**Code**]
- Guangyi Zhang*, Xiao Fu*, Qiyu Hu†, Yunlong Cai, Guanding Yu. "**Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network**". *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Helsinki, Finland, 2021, pp. 678-683 [**Paper**|**Slides**]

Honors & Awards

- National Scholarship, Ministry of Education of P.R. China **(Top 0.2% across China, twice)** *2020,2021*
- ISEE Excellent Student Award **(Top 5 students among ISEE 314 undergraduates)** *2021*
- ISEE Yuelun Alumni Scholarship **(Dean's Award for Academic Contest, Bonus 20,000 CNY)** *2020*
- First-Class Scholarship for Academic Excellence of Zhejiang Univeristy **(Top 3%)** *2020,2021*
- Government Scholarship for Academic Excellence of Zhejiang Province **(Top 3%)** *2019*
- Outstanding Graduation Thesis Award of Zhejiang University *2022*
- Finalist Winner, Mathematical Contest in Modeling (MCM/ICM) **(Top 2% out of 20,954 teams)** *2021*
- Meritorious Winner, Mathematical Contest in Modeling (MCM/ICM) **(Top 7% out of 26,112 teams)** *2020*
- Gold Prize, 7^{th} "Inernet Plus" College Student Innovation and Entrepreneurship Contest **(1^{st} author)** *2021*
- Gold Prize, 12^{th} "Challenge Cup" National College Student Business Plan Competition, National *2020*
- Grand Prize, 12^{th} "Challenge Cup" National College Student Business Plan Competition, Provincial *2020*
- Grand Prize, "Yongdian Cup" Innovation and Entrepreneurship Contest **(1^{st} author)** *2019*
- Second Prize, College Student Physical Innovation Competition (Theoretical) of Zhejiang Province *2020*
- Third Prize, 17^{th} "Challenge Cup" Extracurricular Academic Science and Technology Contest **(1^{st} author)** *2021*
- Outstanding Student Award of Zhejiang University *2019,2020,2021*
- Academic Excellence Scholarship of Zhejiang University *2019,2020,2021*

Research Experience

Large-Scale Panoptic 3D Reconstruction of Real-World Scenario Pujiang National Laboratory
Mentor: Prof. Dahua Lin and Dr. Jiaqi Wang *Aug. 2022 – Now*

- generated high-quality scene reconstruction and panoptic labels in 3D space from multi-view images and semantic label results predicted by 2D pre-trained network on the ScanNet dataset
- leveraged the coarse semantic segmentations of 2D images and the SDFs of scene reconstruction to obtain the accurate instance labels in 3D space, especially in overlapped regions of objects
- employed the reconstruction priors to improve the accuracy of 3D semantic/instance segmentation masks
- intending to publish a pertinent paper to CVPR'23 in capacity of the first author

Unaligned Sparse Object-Centric Image Synthesis Shanghai Artificial Intelligence Laboratory
Mentor: Dr. Jiaqi Wang and Dr. Sida Peng *Mar. 2022 – Jul. 2022*

- focused on view synthesis from sparse inputs on unaligned coordinate system, a more generic scenario, where category-level priors can not be stored by *xyz* inputs
- projected the features of source views in 3D space via monocular depth predicted by MiDaS and replaced *xyz*, from which network can implicitly learn priors, in NeRFormer with nearest features at novel viewpoints
- outperformed NeRFormer w/o *xyz* and ViewFormer in terms of novel view synthesis, and was comparable with NeRFormer in case of sparse views (1-3 views)

Panoptic NeRF: 3D-to-2D Panoptic Label Transfer ZJU
Mentor: Prof. Andreas Geiger and Prof. Yiyi Liao *Sept. 2021 – Mar. 2022*

- proposed to utilize neural rendering as a tool to generate high-quality 2D panoptic segmentation annotations from a set of sparse images and easy-to-obtain coarse 3D bounding primitives and noisy 2D semantic predictions
- leveraged a novel dual formulation of the semantic fields to improve the underlying geometry reconstruction and resolve label ambiguity at the intersection region of the 3D bounding primitives
- reduced the label annotation time from 1.5 hours to 0.75 minute per frame in a single street scene
- achieved SOTA label transfer results in terms of accuracy and multi-view consistency on challenging urban scenes of the KITTI-360 dataset and enabled the rendering of multi-view and spatio-temporally consistent labels at novel viewpoints

Algorithm-Based Dual-Layer Deep-Unfolding Neural Network ZJU
Mentor: Prof. Guanding Yu *Jun. 2020 – Feb. 2021*

- proposed a learning-based framework, Dual-Layer Deep-Unfolding Neural Network (DLDUNN), to address the high computational complexity problem of a convex optimization algorithm, i.e. the dual-layer iterative Penality Dual Decomposition (PDD) algorithm
- unfolded the PDD algorithm into a layer-wise structure whereby some trainable parameters were introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms
- presented high performance of the PDD algorithm ($\geq 97\%$) with a greatly reduced complexity, i.e. total number of iterations (10,000$\rightarrow$70), optimization of $\mathbf{V}_{RF}$ ($\mathcal{O}\left(N^2 N_{RF}^2\right) \rightarrow \mathcal{O}\left(1\right)$) and $\mathbf{U}_{RF}$ ($\mathcal{O}\left(KM^2 M_{RF}^2\right) \rightarrow \mathcal{O}\left(1\right)$), matrix inversion $\rightarrow$ matrix multiplication ($\mathcal{O}\left(N^3\right) \rightarrow \mathcal{O}\left(N^{2.37}\right)$)
- focused on solving the spectrum efficiency maximization problem for hybrid pre-coding architecture
- **Laureate of Excellent Award (Team leader) - Student Research Training Program (Province-level)**

Extracurricular Activities

ISEE Youth League Committee Hangzhou, China
Deputy Secretary *Sept. 2020 – Jul. 2021*

- assisted the league committee teachers in carrying out academic competitions as well as innovation and entrepreneurship competitions ("Internet Plus" and "Challenge Cup")
- participated in the sharing session of outstanding seniors of ISEE, and the counterparts of the 19 grade assembly, epidemic prevention, and employment quality analysis
- led six teams in innovation and entrepreneurship competitions as a consulting assistant

13^{th} **ZJU Morning-side Cultural China Scholars Program** Hangzhou, China
Team Leader *Jun. 2021 – Jun. 2022*

- assisted in the completion of courses given by distinguished intellects from various world-class social communities
- responsible for the 11^{th} Annual Forum which explored the progress and limit of modern China's transformation
- organized overseas investigations, including winter break visits to Hong Kong and Macau, and the summer ones to the United States (Bay area, Chicago, Boston, New York)
- organized 13 issues of the "People in the MCCSP" sharing series, which aimed at sharing and broadcasting Cultural China's alumni's valuable life trajectories, and have gained over 10,000 reads on WeChat platform

ZJU Daily Volunteer Hangzhou, China
Participant *Sept. 2018 – Jun. 2022*

- contributed with an over 250 volunteer hours by participating in more than 30 volunteer services, such as the Chunlin teaching activities, Prince Bay volunteer service, "Internet Plus" reception service, ZJU autumn sporting events, and hometown anti-epidemic activities
- **Laureate of ZJU Five-Star Volunteer, the highest honor awarded for ZJU volunteer participants**

Computer Skills

Python, PyTorch, TensorFlow, C/C++, MATLAB, Java, LaTex, Linux, Caffe.